

Program Obfuscation — iO Powered Proofs

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Program Obfuscation

Definition 1 (Barak et al.)

A virtual black box (VBB) obfuscator is a PPT alg. $\mathcal{O}$ s.t.

1. (functionality) $\forall$ program $P, \mathcal{O}(P) \equiv P$.
2. (polynomial slowdown) $\exists$ poly s.t. $|\mathcal{O}(P)| \leq \text{poly}(|P|)$.
3. (VBB property) $\forall$ PPT $\mathcal{A}$, $\exists$ PPT $\mathcal{S}$ and negl s.t. $\forall P$,

$$|\Pr[\mathcal{A}(\mathcal{O}(P)) = 1] - \Pr[\mathcal{S}^P(1^{|P|}) = 1]| \leq \text{negl}(|P|).$$

Intuitively, $\{\mathcal{A}(\mathcal{O}(P))\} \approx \{\mathcal{S}^P(1^{|P|})\}$.

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→ Turing Machine (M) vs Circuit (C) distinction (property 2).

Unobfuscatable function ensemble

Definition 2 (Barak et al.)

$\{\mathcal{H}_k\}_{k \in \mathbb{N}}$ distributions on eff. comp. finite functions ($f \leftarrow \mathcal{H}_k$) s.t.

$$\exists \pi: \bigcup_{k \in \mathbb{N}} \text{Supp}(\mathcal{H}_k) \rightarrow \{0, 1\}$$

- $\forall$ PPT $\mathcal{S}$,

$$\Pr_{f \leftarrow \mathcal{H}_k} [\mathcal{S}^f(1^k) = \pi(f)] = \frac{1}{2} + \text{negl}(k)$$

- $\exists$ PPT $\mathcal{A}$ s.t.

$$\begin{cases} \forall f \in \bigcup_{k \in \mathbb{N}} \text{Supp}(\mathcal{H}_k) \\ \forall \mathcal{C}: [\mathcal{C}] = f \end{cases} \implies \mathcal{A}(\mathcal{C}) = \pi(f)$$

Barak et al.'s Impossibility Result

“The essence [...] is that there is a fundamental difference between getting black-box access to a function and getting a program that computes it, no matter how obfuscated: A program is a succinct description of the function, on which one can perform computations (or run other programs).”

- Goal: Construct a function that cannot be learned with oracle queries but still leaks information.

Barak et al.'s Impossibility Result

Theorem 3

TM (resp. circuit) VBB obfuscators don't exist.

proof sketch.

Suppose $\mathcal{O}$ is a VBB obfuscator. For $\alpha, \beta \in \{0, 1\}^k$ let

$$C_{\alpha, \beta}(x) = \begin{cases} \beta & x = \alpha \\ 0^k & - \end{cases}, \quad D_{\alpha, \beta}(C_{\cdot, \cdot}) = \begin{cases} 1 & C(\alpha) = \beta \\ 0 & - \end{cases}$$

$$\mathcal{A}(\mathcal{O}(C_{\alpha, \beta}), \mathcal{O}(D_{\alpha, \beta})) \rightarrow \mathcal{O}(D_{\alpha, \beta})(\mathcal{O}(C_{\alpha, \beta})) = D_{\alpha, \beta}(C_{\alpha, \beta})$$

$$\implies \Pr[\mathcal{A}(\mathcal{O}(C_{\alpha, \beta}), \mathcal{O}(D_{\alpha, \beta})) = 1] = \Pr[\mathcal{A}(\mathcal{O}(C_{\alpha, \beta} \# D_{\alpha, \beta})) = 1] = 1$$



1. (Functionality) $\forall$ program P , $i\mathcal{O}(P) \equiv P$.
2. (Indistinguishability) $\forall P_0 \equiv P_1$,
 $|\Pr[\mathcal{A}(i\mathcal{O}(P_0)) = 1] - \Pr[\mathcal{A}(i\mathcal{O}(P_1)) = 1]| \leq \text{negl}.$

Puncturable PRF (PPRF)

Definition 4 (Sahai & Waters)

A PPRF is a bundle $(\text{Key}_F, \text{Puncture}_F, \text{Eval}_F)$ s.t.

1. (Functionality) $\forall$ PPT $\mathcal{A}$ s.t. $\mathcal{A}(1^\lambda) \rightarrow S \subset \{0, 1\}^{n(\lambda)}$,
 $\forall x \in \{0, 1\}^{n(\lambda)} \setminus S$,

$$\Pr \left[\text{Eval}_F(K, x) = \text{Eval}_F(K_S, x) \mid \begin{array}{l} K \leftarrow \text{Key}_F(1^\lambda) \\ K_S \leftarrow \text{Puncture}_F(K, S) \end{array} \right] = 1$$

2. (Pseudorandomness at punctured points) $\forall$ PPT
 $(\mathcal{A}_1, \mathcal{A}_2): \mathcal{A}_1(1^\lambda) \rightarrow (S \subset \{0, 1\}^{n(\lambda)}, \sigma)$,

$$\begin{aligned} & \left| \Pr [\mathcal{A}_2(\sigma, K_S, S, \text{Eval}_F(K, S)) = 1] \right. \\ & \left. - \Pr [\mathcal{A}_2(\sigma, K_S, S, \mathcal{U}_{m(\lambda) \cdot |S|}) = 1] \right| = \text{negl}(\lambda) \end{aligned}$$

• PPRF allows a secret key K to be transformed into a punctured key $K\{x^*\}$ that computes $F(K, x) \forall x \neq x^*$ without revealing information about $F(K, x^*)$.

Constructions

- private-key encryption, PPRF and $\text{iO} \implies \text{PKE}$
- $\text{PPRF} + \text{iO} \implies \text{Deniable Encryption}$

(Deniable Encryption): Given $c = E(m, r)$, a sender forced to reveal (m, r) to $\mathcal{A}$ should be able to provide (m', r') s.t. $c = E(m', r')$.

Practicality?

“An obfuscation of the Advanced Encryption Standard encryption circuit at a security level of 128 bits would measure 18 PB and have an evaluation time of about 272 years” (COSIC, 2020)